

Joseph M. Dalton

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Expected May 2027

Bachelor of Science, Mechanical Engineering

Cumulative GPA: 3.9, Dean's List all semesters,

Relevant courses: Mechanics of Materials, Mechanical Design, Dynamics, Fluid Mechanics

RELEVANT EXPERIENCE

Cornell Electric Vehicle

Sept 2025-Present

Chassis Subteam Member

- Collaborated with 80+ engineers across subsystems, fabricated a carbon fiber chassis, contributing to a 20.5 mi/kWh autonomous electric vehicle that competed in the Shell Eco-Marathon.
- Utilized ANSYS Fluent (Computational Fluid Dynamics) to iterate aerodynamics design, reducing drag by 4% over previous years
- Executed carbon-fiber layups: mold prep, Duratek spray, resin ratios, and precise fiber placement
- Prepared and sanded molds to exact tolerances, yielding smoother surface finish and stronger fiber bonds

Cornell CSalt

Sept 2024-Nov 2025

Membrane Engineer

- Utilized research experience to procure a reverse osmosis membrane and pressure vessel, ensuring these critical components met desired design and efficiency goals.
- In charge of procuring, testing, & integration of carbon pre-filter to ensure no pollutants reached membrane.
- Researched and designed innovative wave-powered desalination device, winning 1st place in the 2025 Marine Energy Collegiate Competition for technical design.

Undergraduate Teaching Assistant—Thermodynamics

Aug 2025-Dec 2025

Cornell University, Ithaca, NY

- Guided students through problems in areas such as energy balances, cycle analysis, and evaluating properties during office hours and review sessions
- Facilitated weekly discussion sections for students to reinforce concepts learned in lecture
- Prepared coursework including quizzes, discussion sections, and exams to enhance course materials, improve student understanding.

ADDITIONAL SKILLS

Microsoft Word and PowerPoint; Adobe Photoshop.

Experience in coding such as MATLAB and Java.

Experience in AutoCAD, Fusion360, and ANSYS.

Machine shop certified.

CAMPUS INVOLVEMENT

- **Cornell University Car Club**, Member: Attended club meetings and gained valuable ownership insight from car owners.
- **Cornell University Men's Club Swim**, Captain: Provided leadership to team for 2024 season.